Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI-Native Internal Tools · MCP Server · Anthropic Claude API · Python · TypeScript · Enterprise Integrations · Agentic Workflows

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SUMMARY

Senior Software Engineer with 8+ years building production backend services, enterprise API integrations, and - in the past two years - AI-native internal tools. Deployed a **custom MCP (Model Context Protocol) server** connecting agentic pipeline outputs to downstream enterprise systems: authentication, structured data access, failure routing, and observability across the integration boundary. Built with the **Anthropic Claude API** (user's daily LLM) - prompt design, structured outputs, tool use, streaming, and response parsing for production agentic workflows. Built **Leasel**, an internal AI platform with three user-facing surfaces (document intelligence, support investigation, self-service) sharing one knowledge repository - HR-adjacent in scope: intake, triage, policy lookup, eligibility routing, and human escalation. At Entrata (8 years, 100K+ users), owned enterprise SaaS integrations, REST APIs with OAuth and rate-limit handling, compliance data pipelines, and internal tooling in a regulated multi-tenant environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

MCP Server Architecture | Anthropic Claude API | Agentic Workflows | Prompt Design · Structured Outputs | Python · FastAPI · Django | TypeScript · Node.js | REST APIs · OAuth · Rate Limiting | Enterprise SaaS Integrations | LangGraph · LangChain | RAG · Hybrid Search | HITL · Audit Logging | PostgreSQL · SQLite · Redis | AWS · Azure · GCP | Docker · Kubernetes | CI/CD · GitHub Actions | Internal Tooling | Compliance & Regulated Data | PHI · HIPAA · SOC 2

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data pipelines, and Python API layers connecting AI components to backend services; debugging data-driven and AI-powered features across the full stack.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- **Deployed a custom MCP (Model Context Protocol) server** giving AI agents structured access to downstream enterprise system data - defined the tool interface, handled OAuth-style authentication, built failure routing and observability, and enforced access boundaries so agents could only invoke explicitly permitted operations.
- Built an agentic mission-planning system (LangGraph StateGraph) with RAG over regulatory documents - natural-language input to structured outputs via the Anthropic Claude API; validation guardrails blocking non-compliant actions before tool invocation and export.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming endpoint pipeline from regulatory corpus to real-time operator interface; integrated with enterprise tooling via defined API contracts.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ users, multi-tenant)

- **Owned enterprise SaaS integrations and REST API contracts** across a multi-tenant platform serving 100K+ users - OAuth flows, rate-limit handling, pagination, retry semantics, error handling, and long-term API contract maintenance; third-party integrations with defined input/output contracts and acceptance criteria.
- **Built and maintained internal tooling** that reduced engineering debugging time by 40% and kept bug counts below 10 for 6+ consecutive months; automated operational workflows eliminating 80% of manual workload via Python/Bash pipelines.
- **Owned compliance data pipelines in a regulated environment** - HUD income-limit import (few thousand federal rows → ~180,000 domain records), eligibility and rent calculations for 100K+ housing units; 7 years of zero compliance violations; PHI-encrypted pipelines (Libsodium), 100% coverage, zero security incidents.
- Built RabbitMQ-based async distributed processing with retry logic and idempotency; optimized 200+ SQL queries (30% gain); shipped 20+ full-stack modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints.
- Mentored 5+ engineers; conducted 300+ peer code reviews; led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built backend modules (Python, PHP, Java, PostgreSQL) including large-scale data imports, document management, and compliance workflows; led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - AI-Native Internal Platform (Document Intelligence + Agentic Workflows)

2026 | Python, LangGraph (8-agent), FastAPI, Anthropic Claude API, SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Three internal surfaces sharing one knowledge repository: **document intelligence** (OCR, hybrid search, citation grounding, clause detection), **support investigation** (8-agent LangGraph pipeline: intake → triage → retrieval → configuration lookup → historical incident → structured reasoning → validation → human escalation), and **self-service** with a hard eligibility boundary - policy questions answered directly, regulated decisions always route to staff.

- Built with the **Anthropic Claude API** - prompt design, structured outputs, tool use, and confidence-gated HITL escalation; audit logging on every agent decision; CI regression harness with LLM-as-judge scoring; 93.6% precision across 1,240 test cases.

AgentProbe - Open-Source LLM Evaluation Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; LLM routing across Anthropic, OpenAI, Azure OpenAI, Groq, and Ollama with rate-limit handling and response parsing; 5-metric eval engine; CI-integrated regression harness with golden datasets and structured logging for internal tool quality verification.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017